

Indraneel Samanta

LinkedIn | GitHub | Portfolio

Education

Dwarkadas J. Sanghvi College of Engineering (DJSCE)
B.Tech in Artificial Intelligence & Machine Learning (AIML)

Mumbai, India
2024 – 2028 (Expected)

Skills & Languages

Programming & Databases: Python, SQL (PostgreSQL, SQLite), C++, C, SQLAlchemy

Data Science & Analytics: ETL Pipelines, A/B Testing, Feature Engineering, Statistical Analysis, Exploratory Data Analysis (EDA), Data Visualization (Plotly, Seaborn)

Machine Learning & MLOps: XGBoost, Random Forest, Scikit-Learn, Optuna, MLflow, DVC, Pandas, NumPy

Cloud, Backend & Tools: AWS, Docker, CI/CD (GitHub Actions), FastAPI, REST APIs, Supabase, Pytest, Git

Spoken Languages: English (Fluent), Japanese (Preparing for JLPT N1)

Technical Projects

Spotify ML Pipeline & Data Engineering | AWS, PostgreSQL, XGBoost, FastAPI

- **Optimized dataset footprint by 85% (277MB to 42.5MB)** via algorithmic downcasting, and architected an ETL pipeline utilizing PostgreSQL native buffer streaming to **slash export times from 15 minutes to 3.2s**.
- **Engineered a 19-feature matrix** to predict user skips with **≥80% precision**, employing chronological walk-forward splitting and **GPU-accelerated Optuna** (doubling throughput to 4 it/s compared to CPU-only) for 300-trial XGBoost hyperparameter tuning.
- **Architected a Financial Utility A/B Testing Simulator** to quantify model ROI, simulating **838 MB of CDN bandwidth savings** across the dataset and proving the high-precision strategy maximizes dollar savings by avoiding false-positive user disruptions.
- **Deployed a real-time FastAPI microservice** and interactive queue dashboard using Docker, dispatching live model predictions to a **Dual-Policy Action Engine** for dynamic CDN bandwidth throttling.
- **Enforced strict MLOps and production reliability standards** by orchestrating a 20-test Pytest suite, GitHub Actions CI/CD pipelines, and managing heavy model artifacts via **Data Version Control (DVC)**.

Otakufy | Full-Stack Japanese Learning Platform | PostgreSQL, Next.js, NLP (Kuroshiro)

Live: otakufy.vercel.app

- **Engineered an automated data pipeline to audit and sanitize a 14,082-item JSON dataset**, conducting deep linguistic, conjugation, and redundancy scans to ensure **100% byte-parity** with PostgreSQL seeds.
- **Optimized backend data retrieval** by replacing slow SQL `ORDER BY random()` operations with **CTE indexed slice queries**, enabling the millisecond shuffling of the entire 14,082-item question bank.
- **Integrated Japanese NLP tokenization pipelines (Kuroshiro & Wanakana)** to programmatically parse complex Kanji structures and dynamically generate phonetic reading aids, enabling automated content scaling.
- **Enforced strict data integrity via PostgreSQL Row-Level Security (RLS)** policies, ensuring leaderboard authenticity and preventing unauthorized state mutations across concurrent API queries.

Personal Portfolio & Interactive Resume | React 19, REST APIs, JSON (i18n)

Live: indraneelsamanta.vercel.app

- **Executed comprehensive cybersecurity audits and static code analysis**, successfully remediating high-severity CWE vulnerabilities (Path Traversal, DoS) and verifying XSS DOM sanitization.
- **Integrated a production-ready REST API pipeline for automated contact routing**, implementing strict client-side honeypot mechanisms and input sanitization to mitigate automated bot spam.

Experience

Logistics & Events Co-committee | Google Developer Student Club (GDSC) DJSCE

Oct 2025 – Present

- **Led a 5-member operations team to manage and execute** end-to-end logistics for collegiate hackathons, successfully coordinating venue scheduling, technical infrastructure, and event-day operations to support **350+ attendees**.
- **Managed corporate outreach and sponsor communications**, pitching technical event roadmaps to industry partners to successfully secure funding, while driving participant registration across multiple campus networks.